



Automatic lung disease classification from the chest X-ray images using hybrid deep learning algorithm

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Received: 4 April 2022 / Revised: 30 August 2022 / Accepted: 27 February 2023

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Abstract

The chest X-ray images provide vital information about the congestion cost-effectively. We propose a novel Hybrid Deep Learning Algorithm (HDLA) framework for automatic lung disease classification from chest X-ray images. The model consists of steps including pre-processing of chest X-ray images, automatic feature extraction, and detection. In a pre-processing step, our goal is to improve the quality of raw chest X-ray images using the combination of optimal filtering without data loss. The robust Convolutional Neural Network (CNN) is proposed using the pre-trained model for automatic lung feature extraction. We employed the 2D CNN model for the optimum feature extraction in minimum time and space requirements. The proposed 2D CNN model ensures robust feature learning with highly efficient 1D feature estimation from the input pre-processed image. As the extracted 1D features have suffered from significant scale variations, we optimized them using min-max scaling. We classify the CNN features using the different machine learning classifiers such as AdaBoost, Support Vector Machine (SVM), Random Forest (RM), Backpropagation Neural Network (BNN), and Deep Neural Network (DNN). The experimental results claim that the proposed model improves the overall accuracy by 3.1% and reduces the computational complexity by 16.91% compared to state-of-the-art methods.

Keywords Computer aided diagnosis · Convolutional neural network · Deep learning · Features scaling · Lung disease · X-ray image

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1 Introduction

Lung diseases carried about by different components have prompted a higher death rate over the most recent couple of years. The people groups infected from novel Covid-19 and pneumonia have moderate or gentle side effects like fever, hacking, and inhaling brevity [70]. Nonetheless, certain individuals experienced extreme pneumonic conditions in their lungs that brought about death likewise [25, 59, 68]. A large portion of the cases that kicked the bucket from Coronavirus had experienced high chest blockage (Pneumonia) as a huge decrease in oxygen level and subsequently significant cardiovascular failure. On the opposite side, Pneumonia is likewise a sort of lung sickness that prompts irritation in the little air sacs inside the lungs of the human body. It might top off a lot of liquid which makes it hard to relax. Pneumonia can be brought about by different reasons like viral contaminations (like Covid-19, bacterial influenza, or viral pipe), normal cold, and bacterial diseases [41]. Because of the appearance of Covid-19 illness, it is an extremely moving assignment for clinical specialists to distinguish lung diseases (either popular/bacterial pneumonia or Covid-10 pneumonia) from chest X-ray images [82]. Subsequently, our concentration in this article is the early recognition and grouping of lung illnesses from crude X-ray images for suitable treatment to decrease the death rate brought about by high chest clogs [60]. The lung disease brought about by a novel Covid is called Novel Coronavirus Infected Pneumonia (NCIP).

An alternate sort of lung infection that has a huge danger to people is lung disease [72]. The World Health Organization (WHO) asserts that roughly 8 million individuals experienced lung malignant growth [84]. In any case, these are not huge numbers considering the number of years that guarantee 8 million lung malignant growth patients contrasted with lung sicknesses caused by Covid-19 and Pneumonia within the brief time frame. To fix lung malignant growth, a few investigations have effectively introduced its initial expectation by utilizing computer vision procedures and delicate processing strategies [44, 46, 50]. The location of lung disease has been performed utilizing methods like X-ray, Magnetic Resonance Imaging (MRI), Computed Tomography (CT), and isotope. Among these, CT and X-ray chest imaging strategies are much of the time utilized for the recognition of different lung illnesses. X-ray and CT images are used by radiologists and doctors to find lung sicknesses. Among the CT sweep and X-ray, the X-ray strategy is practical with comparative sorts of results contrasted with the CT check. Thus, many specialists suggested the Chest X-ray for the examination of lung infections, particularly during the Covid-19 period. The X-ray method has been utilized to analyze irregularities in the areas of the human body like the chest, skull, bones, teeth, and so on for a long time, clinical specialists have utilized the X-ray strategy to dissect and investigate the different irregularities in the human body organs [42]. Many investigations uncovered that X-rays have a practical strategy for sickness diagnosis while uncovering the obsessive changes alongside their monetary effectiveness and non-obtrusive properties [80]. The lung diseases can be addressed in chest X-ray images in the type of solidifications, dulled costophrenic points, extensively disseminated knobs, cavitations, and invades [4]. In the investigation of the X-ray image of the patient, radiologists recognize a few conditions like pneumonia, nodule, pleurisy, radiation, invasion, fractures, pneumothorax, pericarditis, and so on [54, 81].

The discovery and arrangement of lung sicknesses utilizing chest X-ray images are viewed as an intricate interaction for radiologists, along these lines, it got huge consideration from the scientists for programmed lung infection identification. Since the previous decade, numerous Computer-Aided Diagnosis (CAD) frameworks presented utilizing X-ray images for diagnosis

purposes. Yet, such frameworks neglected to accomplish the necessary presentation for lung disease discovery and orders [61]. The new Covid-19 helped lung diseases further making the assignments exceptionally trying for such CAD frameworks as it is fundamental to distinguish the presence of pneumonia in the lungs and its arrangement to either Covid-19, bacterial, or viral contaminations. This arrangement assists with concentrating on pneumonic patients. Since the event of Covid-19, a lot of work has been introduced as a CAD framework for Coronavirus and pneumonia sickness locations utilizing chest X-ray images utilizing robotized image handling and deep learning strategies [26]. As deep learning is a mechanized element learning and extraction procedure, it sets aside a more drawn-out effort for complete dataset preparation and identification purposes. Furthermore, thus, such arrangements are not solid and hearty against the expanded number of datasets. Deep learning procedures like Convolutional Neural Network (CNN) acquired critical consideration for lung sickness identification because of their capacity to improve precision and programmed feature extraction [10]. But such methods are suffering from the severe challenges of high processing time and space requirements during the automatic features learning and estimation from the input images. Apart from this, the existing deep learning-based solutions heavily relied on the automatically extracted features from the 3D input images that contain significant variations and hence suffer from the challenges like vanishing gradient explosion and higher miss classification rates. The current approaches delegate the task of both feature extraction and classification to the CNNs which is another reason for higher space and time.

We proposed a novel deep learning model for the automatic classification of lung diseases from chest X-ray images called HDLA. The new aspect of this HDLA model is that it utilizes a streamlined approach for the extraction of features and the independent categorization of those features. We chose to create the 2D CNN layers with optimal feature size rather than the 3D model to cut down on the amount of processing time required for the automated extraction of CNN features. We have pre-processed the raw X-ray images by using the most effective strategy to estimate reliable characteristics for each chest X-ray image. To overcome the challenges of the vanishing gradient and higher computational efforts, we applied different classifiers to the CNN outcomes for lung disease classification. The main practical implication of this research is to control or monitor lung-related diseases from chest X-ray scans. The earlier prediction of lung diseases like pneumonia or covid-19 correctly will help to reduce the mortality rate. Since the arrival of Covid-19, it becomes challenging to distinguish between pneumonia and Covid-19 infections. We have studied the various recently proposed image processing and deep learning models [14, 23, 24, 33, 43, 51, 53, 71, 76] for the healthcare systems [3, 48] for the practical implications of the present study. Apart from this, we have studied the Artificial Intelligence (AI) based works [15, 16, 19] in the field of pattern recognition during this research as well. The various other methods that have recently been proposed for breast cancer detection and heart disease prediction have been presented using image processing, signal processing, and classification techniques [11, 12, 29, 30, 34, 35]. The contributions of the proposed model are briefly described as: (1) The hybrid CAD system recommended improved lung disease classification utilizing chest X-rays by using pre-processing, robust CNN, feature scaling, and classifiers, (2) We employed a lightweight but successful pre-processing approach to enhance chest X-ray images before feeding them to CNN for feature extraction, (3) Robust 2D CNN is described for efficient feature extraction using the pre-trained ResNet50 model. Min-max scaling improves 2D CNN output, and (4) We employed Adaboost, RM, SVM, BNN, and DNN to classify chest X-ray images with vanishing gradients. The remainder of the paper consists of sections such as the related works

that have been studied in section 2, section 3 presents the methodology of the proposed model, section 4 presents the simulation results and discussions, and section 5 shows the conclusion and recommendations for future work.

2 Related works

Over the last few years, lung disease classification has received vital interest using deep learning and machine learning mechanisms. Deep learning and machine learning methods have been combined with different computer vision techniques [17, 18]. Recently several works have been proposed on Covid-19 prediction using machine learning and deep learning models [9, 36–38, 52, 64, 66]. We present a study of the recently proposed studies for automatic lung disease classification using chest X-ray images in this section. According to the recent works, we described the research gaps and proposed contributions.

2.1 Automatic state-of-art methods

In [2], authors have developed a deep learning-based system for identifying chest infections using chest X-rays. Using an X-ray dataset, they planned and tested an automated CNN model for diagnosing chest pain. In terms of preparing precision, testing exactness, and preparation time, the creator discovered significant execution aftereffects of the CNN model with other sensitive registering procedures. In [75], the authors have used computer vision methodologies and careful registration techniques to detect pneumonia in chest X-ray images. The Region of Interest (ROI) was extracted by dividing X-ray images, after which surface provisions were eliminated and then applied to the neuronal organizations for grouping. In [47], CNN planned for chest X-ray scans to be used to detect pneumonia. They created the dataset from the Kaggle archive and designed ConvNet to handle the information X-ray image. The lung illness identification method makes use of computer vision methods and the useful CNN model introduced in [77]. The division computation had used to determine ROI in lung images, and then the neighboring and global provisions were deleted for viable pneumonia grouping. The amiable CNN model had planned to act out the grouping. Another deep learning-based technique was recently published in [73], where the author used a CNN model named VGG16 to classify pneumonia using an X-ray chest image dataset. During the learning stage, they used the exchange learning and modifying technique. Method for detecting pneumonia using X-ray images and CNNs had proposed in [5]. They programmed the CNN to classify the information in the X-ray image as ordinary or pneumonic. The precise and effective pneumonia detection using the information chest X-ray image had proposed in [67]. They began by pre-preparing the information X-ray image using appropriate filtering and distinction enhancement approaches. They build deep leftover learning using different convolutional networks for grouping. Another method for locating pneumonia infections based on CNN had given in [56]. They prepared X-ray images of common and unusual illnesses and built a model to detect the presence of pneumonia. The novel solution employing the weighted delicate figuring technique was introduced in [31] utilizing weighted expectations from common deep learning frameworks such as DenseNet121, MobileNetV3, Xception, and ResNet. To predict results based on dataset quality, a managed learning system had presented. The approach for locating pneumonia from chest X-ray input images had proposed in [28]. They planned to use CheXNet, a deep CNN model, in conjunction with VGG-19 for highlight extraction. For

grouping, the components were assembled. To address the issue of information anomaly, they offered solutions such as Synthetic Minority Oversampling Technique (SMOTE), Random Over Sampler (ROS), and Random Under Sample (RUS).

To play out the characterization of Covid-19 illness using X-ray images, the deep learning model developed in [1] was dubbed Decompose, Transfer, and Compose (DeTraC). The DeTraC approach proved effective in dealing with information anomalies. CNN's were planned again in [22] for pneumonia grouping based on the transformation of VGG-19, choice tree, and Inception V2 over CT filter images and X-ray images. A programmed method for detecting and ordering Covid illnesses had proposed in [7]. They assemble a dataset of conventional and Covid-19 participants by collecting chest X-ray images. They developed and tested a CNN model for programmed infection forecasting. COVIDDetectionNet, the master-planned model, was suggested in [74] for the characterization of Covid-19 from chest X-ray images. They made use of supplies from a variety of deep components. They used a pre-prepared CNN-assisted AlexNet model in conjunction with a transfer learning method. They aid in highlighting the determination approach that had been familiarised with selecting the strong components from each of the layers of deep learning design. The delicate registering approach SVM was then used for grouping at that moment. A technique for recognizing and categorizing Covid-19 illness into bacterial pneumonia, viral pneumonia, and the typical class had presented in [32]. They used a deep transfer learning technique to apply the concept to multiple chest X-ray datasets of varying sizes. In [27], two ensemble deep transfer learning frameworks were designed for detecting Covid-19 infections using a chest X-ray image. They used the pre-prepared models to improve recognition performance. They pretended to be Coronavirus, bacterial pneumonia, and viral pneumonia.

The unique CAAD (Confidence aware Anomaly Detection) had been presented in [83] for the detection of pneumonia utilizing chest X-ray images. The CAAD model included a common feature extractor that used deep learning, anomaly detection, and confidence prediction. CheXGCN had proposed in [20] for the categorization of chest X-ray images using GCN (Graph Convolution Networks). The CheXGCN model was divided into two phases: IFE (Image Feature Embedding) and LCL (Label Co-occurrence Learning). The innovative CSEN (Convolutional Support Estimation Network) had been presented in [79] to solve the limitations of execution speed and space. The CSEN proposes bridging the neural network and representation-based mechanisms gap. They used the CheXNet pre-trained CNN model for automated feature extraction, which was then followed by feature normalization and machine learning classifiers. Another transfer learning-based automated approach for lung disease (Covid-19) categorization utilizing chest X-ray images had proposed in [58]. They create multiple CNN architectures by utilizing the pre-trained ImageNet model for automated feature extraction from X-ray images. For categorization, CNNs were integrated with several machine learning approaches. In this study, we dubbed our technique CNN-TL (CNN with Transfer Learning). In [57], the CAD method had presented for classifying input chest X-ray images into lung illnesses (non-Covid-19 pneumonia or Covid-19 pneumonia) or healthy classes. They created a CNN model for automated feature extraction and classification utilizing the pre-trained VGG16 model. As a result, it was dubbed the VGG16 Based Model (VGG16-BM). Another significant work [45], advocated automated pneumonia detection from chest X-ray images. They employed a deep transfer learning technique and proposed Weighted Ensemble CNNs, an ensemble of three CNNs utilizing the weighted average ensemble approach (WE-CNNs). DenseNet-121, ResNet-18, and GoogLeNet CNN models were

employed. Aside from the approaches described above, we examined several other publications on lung disease categorization in [6, 8, 13, 21, 49, 62, 65, 69] using various methodologies.

2.2 Motivation

We reviewed methods that mainly used automatic feature extraction techniques using the deep learning CNN models. Most of the techniques employed the transfer learning mechanism using the pre-trained models. All the above methods [1, 2, 5–8, 13, 20–22, 27, 28, 31, 32, 45, 47, 49, 56–58, 62, 65, 67, 69, 73–75, 77, 79, 83] were focused on automatic lung disease classification using chest X-ray images. Despite promising outcomes of such CAD systems, some challenges are still unaddressed for the chest X-ray image-based lung disease classification. We summarize these challenges as:

- None of the state-of-art methods [1, 2, 5–8, 13, 20–22, 27, 28, 31, 32, 45, 47, 49, 56–58, 62, 65, 67, 69, 73–75, 77, 79, 83] focused on image quality enhancement. It limits the reliability of the proposed models as the low-quality X-ray images failed to produce the vital ROI-specific features using the CNN models.
- The 3D CNN models [20, 45, 57, 58, 79, 83] were designed by considering the 3D input for the feature extraction that takes higher processing time and memory space. Using the 3D CNNs for automatic lung disease classification leads to a computationally inefficient CAD system.
- The CNN models produce the high-dimensional features vector that contains significant variations among all the extracted features. Such variations lead to problems like time-consuming training, optimization stuck in local optima, and the worst error surface shape. It also affects classification performance. Only [79] adopted the features scaling, but suffered from other problems.
- The performance analysis of state-of-art studies had performed using the maximum training samples and minimum test samples that limit the scalability of CAD systems.

2.3 Novel contributions

We have proposed a novel automatic CAD system for lung disease classification from the input chest X-ray images called HDLA. As the name suggests, the HDLA functionality has derived from the mechanism of hybrid processing for efficient and robust disease classification. The main contributions of HDLA are as follows.

- Using pre-processing, robust CNN, features scaling, and classifiers, the hybrid CAD system suggested improving the performance of automatic lung disease classification using chest X-ray images.
- To improve the quality of input chest X-ray images before giving them to the CNN for feature extraction, we used a lightweight but effective pre-processing technique.
- Robust 2D CNN is presented for effective feature extraction utilizing the pre-trained ResNet50 model, which employs layers such as a 2D convolutional layer, max-pooling layer, residual blocks, and Global Average Pooling (GAP) with effective kernel sizes for

quicker processing. The min-max features scaling technique is used to better enhance the output of 2D CNN.

- To tackle the vanishing gradients problem while categorizing input chest X-ray images in either of the classes, we used several machine learning algorithms such as Adaboost, RM, SVM, BNN, and DNN.
- The HDLA model is developed and tested using two publicly accessible datasets (Covid-19 Radiography Database (C19RD) [78] and Chest X-Ray Images for Pneumonia (CXIP) (<https://www.kaggle.com/tawsifurrahman/covid19-radiography-database>)), with 70% training and 30% testing using a 10-fold cross-validation approach. The results outperform those of state-of-the-art approaches, demonstrating the system's potential for practical use.

3 Proposed methodology

Figure 1 shows the architecture of the proposed HDLA model for automatic lung disease classification. As per the contributions discussed above, Fig. 1 shows their mechanisms. The proposed architecture has demonstrated both training and testing functions for lung disease detection using the main steps such as pre-processing, CNN features extraction and classification. For quality improvement, each chest X-ray image has first pre-processed using optimal filtering and contrast adjustment techniques. The improved chest X-ray image has further fed to the robust CNN model for the estimation of the features using the pre-trained ResNet50 model. The pre-trained ResNet50 model effectively assists the automatic features learning from the pre-processed chest X-ray images. As shown in Fig. 1, the testing phase shows the

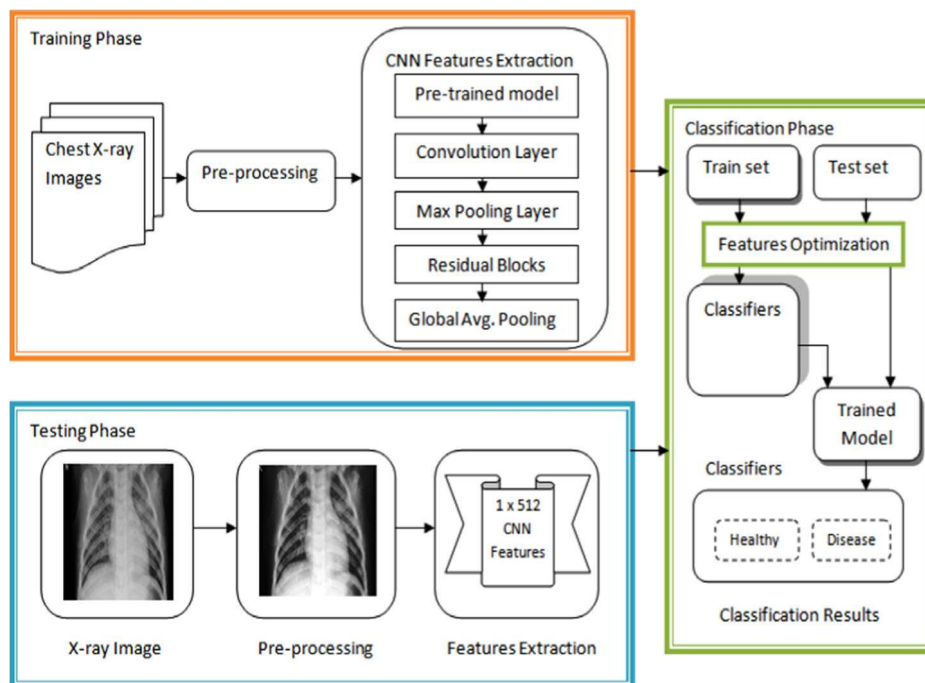


Fig. 1 Proposed functionality of automatic lung disease classification

outcome of the pre-processing step and CNN features extraction. The difference between the original X-ray image and pre-processed X-ray image indicates the impact of applying the pre-processing. The outcome of the proposed CNN model is the features vectors of size 1×512 . After automatic feature extraction, the classification phase has launched to classify the features of input X-ray images in either of the classes. In the classification phase, training and test feature vectors are optimized using the min-max scaling technique to overcome the challenges discussed earlier. After the features optimization, different underlying classifiers have applied to produce their trained models and classification outcome. The designs of each phase are as follows.

3.1 Image quality enhancement

The image quality has a vital factor across different image or video processing applications as the low-quality inputs mislead the outcomes and may result in serious consequences. In the medical domain, advanced biometric scanning tools produce lower-quality medical images like X-ray images. The existing automatic CAD systems of lung disease classification failed to address the quality issues of the X-ray images. Therefore, it limits the reliability of CAD system models to some extent considering the real-time patient monitoring approach. To end this, we tend to improve the quality of the input images by applying suitable and lightweight techniques. First, we standardize each input chest image by transforming it into a grayscale image and resizing it to 512×512 . To balance the trade-off among the image quality improvement with minimum data loss, we applied the three functions such as contrast adjustment, wiener filtering, and histogram equalization on input 2D/grayscale X-ray image x . First, we have applied the contrast adjustment operation to improve the low-quality regions in the input image as:

$$x1 = imadjust(x) \quad (1)$$

After applying the *imadjust* (.) function, we received the contrast improved the chest X-ray image. However, it leads to artifacts and noises in the outcome of the contrast adjustment function. Therefore, we applied the 2D wiener filtering on the $x1$ to produce the filtered image. We tried other filtering techniques, but the wiener filtering produced effective outcomes based on quality metrics Peak to Signal Noise Ratio (PSNR), Structural Similarity Index Matrix (SSIM), and Root Mean Square Error (RMSE). Table 1 shows the average outcomes for PSNR, SSIM, and RMSE using both datasets. The wiener filtering shows better outcomes compared to other techniques as it is an adaptive noise suppression technique. We applied wiener filtering using the default neighborhood size N as [59].

$$x2(i, j) = wiener2\{x1(i, j) | (i, j) \in N\} \quad (2)$$

Table 1 Quality metrics analysis of different filters

Filtering Methods	PSNR	SSIM	RMSE
Average filter	45.91	0.85	113.45
Gaussian filter	43.42	0.81	119.23
Median filter	47.38	0.90	107.34
Bilateral filter	46.12	0.88	111.27
Wiener filter	48.27	0.94	101.23

In *wiener2* function, first, it computes the local variance and mean around each pixel and then applied pixel-wise wiener filtering to produce the filtered image $\times 2$. After that, we focus on improving the final contrast estimation of the $\times 2$ image by applying the histogram equalization function as:

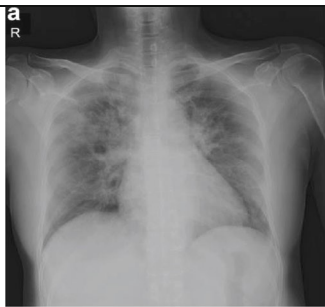
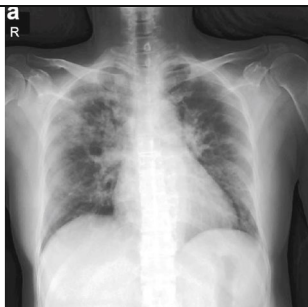
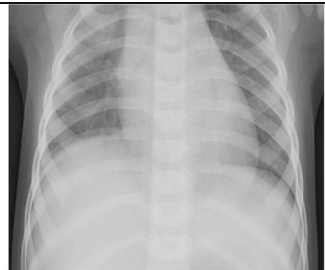
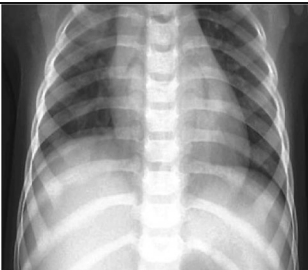
$$x_3 = \text{histeq}(x_2) \quad (3)$$

We required histogram equalization to produce the new map of filtered image $\times 2$ for better quality representations. The benefits of the above steps are justified in Table 2 which shows the test sample and its pre-processed outcome for each dataset.

3.2 Robust CNN model

Considering the 2D pre-processed input chest X-ray image of size 512×512 , we built the 2D CNN model based on the 2D ResNet50 pre-trained model. The ResNet50 is a CNN model with 50 deep layers trained on the ImageNet. The ResNet50 is more powerful for automatic feature extraction with fast processing speed and minimum memory requirements compared to other CNN models. We transformed the input pre-processed image into 224×224 size as the ResNet50 takes the image of the same size as the input. To prevent the challenges of overfitting, faster feature extraction, and facilitating training, we designed 2D CNN model in this paper. The detailed structure of the proposed 2D CNN model is shown in Fig. 2. Table 3 presents the information about each layer configuration.

Table 2 Examples of pre-processing

Dataset	Original Image	Pre-processed Image
C19RD		
CXIP		

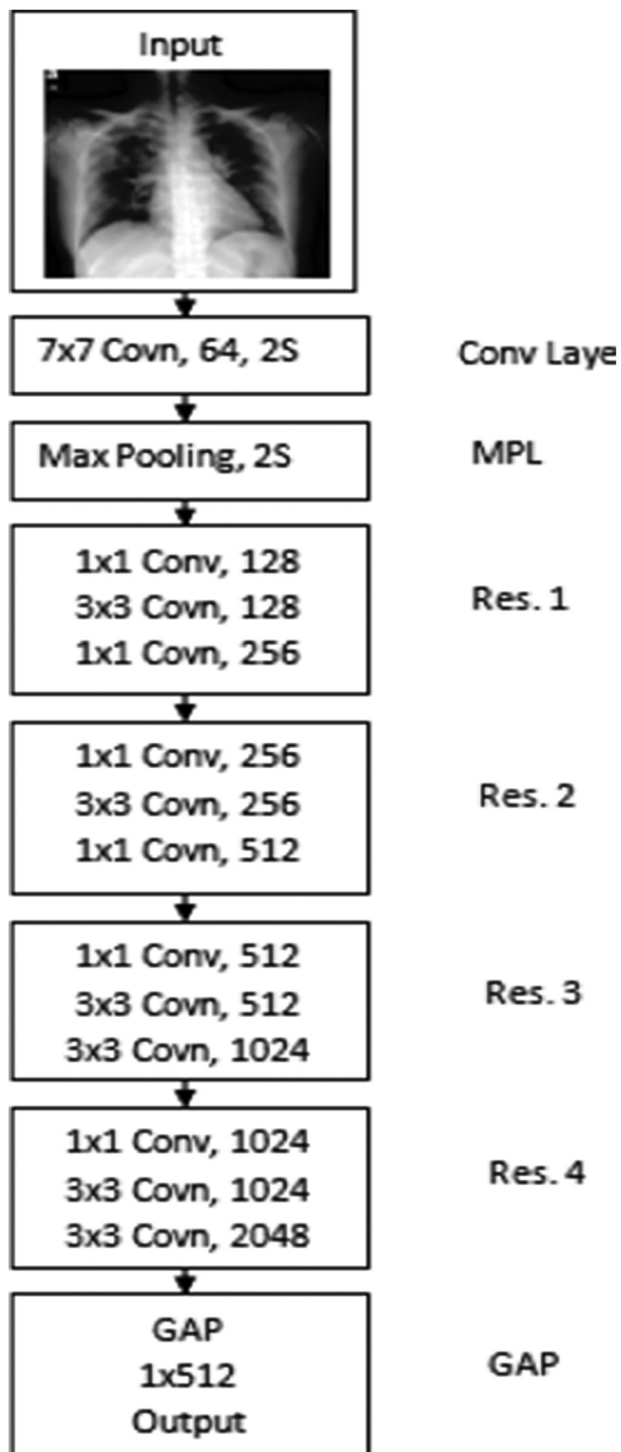


Fig. 2 Structure of proposed 2D CNN model

Table 3 Proposed 2D CNN layers configurations

Layer Name	Output Size	Configurations
Input layer	224×224	Input layer (224, 224)
Conv	112×112	7×7 Conv, 64, Stride 2 (2S)
MPL	112×112	3×3 max pool, 2S
Res 1	56×56	1×1 Conv, 128 →ReLU 3×3 Conv, 128 →ReLU 1×1 Conv, 256 →ReLU
Res 2	28×28	1×1 Conv, 256 →ReLU 3×3 Conv, 256 →ReLU 1×1 Conv, 512 →ReLU
Res 3	14×14	1×1 Conv, 512→ReLU 3×3 Conv, 512 →ReLU 1×1 Conv, 1024 →ReLU
Res 4	7×7	1×1 Conv, 1024 →ReLU 3×3 Conv, 1024 →ReLU 1×1 Conv, 2048 →ReLU
GAP	1×512	Average pooling, 2048/4

As shown in Fig. 2 and Table 3, the proposed CNN model consists of different layers such as the input layer, convolution (Conv) layer, four residual blocks (Res 1 to Res 4), and Global Average Pooling (GAP) layers. At the input layer, we fed the pre-processed chest X-ray image of size 224 × 224 into the network. The Conv layer exploits the diverse kernels to convolve the feature vectors to obtain high-level semiotic data. The max-pooling layer has then applied to minimize the dimensions of Conv layer feature vectors. Every residual block (Res 1 to Res 4) consisted of three blocks rather than two. In the original ResNet-50 model, each residual block always has three stacked convolution layers such as 1 × 1, 3 × 3, and 1 × 1.

The first 1 × 1 Conv has to reduce the feature vectors dimension, the second 3 × 3 Conv has to estimate the feature vectors, and the third 1 × 1 Conv layer has to increase the dimension of feature vectors again. In short, the goal of the 1 × 1 Conv filter is to reduce and increase the size of feature vectors. It helps to solve the degradation problem. We have further optimized these layers by modifying the original Res 1 to Res 4 layers for the faster processing of feature extraction. In the existing ResNet-50, the third 1 × 1 Conv layer increases the feature vector dimensions by four times (e.g., 3 × 3 Conv layer of size 256 feature vector transformed into 1 × 1 1024 feature vector. It leads to high computational requirements in terms of space and time. To reduce the computational requirements, we employed the first layer 1 × 1 in Res 1 set to double (128) of the previous layer, and the third layer 1 × 1 in Res 2 increases double the second 3 × 3 layer (i.e., 128 increased to 256). These modified residual blocks lead to robust network operations with the optimized feature set. Every Conv layer has connected with the batch normalization layer for speedy and table network training. We have dropped the Fully Connected layer as we only need automatically extract features from the input test image. After Res 4 layer, we perform the GAP layer to produce the 1D feature vector of size 1 × 512 from the original size of 1 × 2048.

We further present the mathematical modeling of the proposed CNN model below. As discussed above, the proposed CNN takes input and performs the consolidated one squashing function as per the design of each layer as:

$$x^{2D^l}_j = \tanh\left(\text{pool}_{\max}\left(\text{ReLU}\left(\sum_i y^{l-1}_j(x3)*k_{ij}\right) + b^l_j\right)\right) \quad (4)$$

$$x^{\text{CNN}} = \text{gap}\left(x^{2D^l}_j/4\right) \quad (5)$$

Where,

- $x_j^{2D^l}$ is result of 2D CNN features extraction using convolutional layer l of j^{th} input,
- y_j^{l-1} represents the previous convolutional layer features maps of x_3 ,
- k_{ij} represents i^{th} trained convolutional kernels,
- b_j^l represents the additive bias.
- $\tanh(\cdot)$ represents the activation function,
- $pool_{max}(\cdot)$ represents the operation of max pooling for features extraction,
- $ReLU(\cdot)$ represents the operation of ReLU layer.
- $gap(\cdot)$ represents the global average pooling layer to estimate the final feature vector x^{CNN} .

3.3 Features optimization

As the automatically extracted CNN features contain significant variations, we aim to scale them before applying the classifiers in this paper. The challenges of applying classifiers with the originally extracted CNN features are already discussed earlier in this paper. The features with a greater range perform a definitive role in the training method of classifier algorithms. As the machine learning algorithms concentrated on estimates without knowing their consequence, the features set with distinctions result in incorrect classification outcomes. Also, a raw set of features takes a longer convergence time for neural networks. Therefore, features scaling is required to enhance speed and accuracy. In this paper, we adopted the min-max normalization technique that scales each feature into the range 0 to 1. The min-max normalization applied on x^{CNN} to produce the scaled feature vector x^{scaled} as:

$$x^{scaled} = \frac{(x^{CNN} - \min(x^{CNN}))}{(\max(x^{CNN}) - \min(x^{CNN}))} \quad (6)$$

3.4 Classification

For the classification of the chest X-ray images, we used different classifiers such as Adaboost, SVM, RF, BNN, and DNN. The AdaBoost classifier is the ensemble classifier using the multiclass TotalBoost algorithm. For the multiclass SVM algorithm, we adopted the linear kernels. The RF is another ensemble classifier based on the decision tree approach. The BNN is the neural network classifier and DNN is the deep neural network classifier. We designed DNN only for the classification purpose in this paper. We perform classification using steps:

1. Model training
2. Model testing
3. Repeat 1 and 2.

3.5 Data sources

Each classifier has trained on a 70% training dataset and it is used to classify the 30% test dataset. The training and testing dataset contains the scaled features for each chest X-ray

Table 4 CXIP dataset overall statistics [78]

“Number of chest X-ray samples	5856
Number of normal chest X-ray samples	1341
Number of bacterial pneumonia chest X-ray samples	2890
Number of viral pneumonia chest X-ray samples	1483
Total test samples (30%)	1758
Total training samples (70%)	4098”

image. Moreover, we employed data augmentation in the training set of both datasets. Tables 4 and 5 show the complete statistics for both datasets CXIP [78] and C19RD (<https://www.kaggle.com/tawsifurrahman/covid19-radiography-database>) which consist of total samples, training samples, and testing samples.

4 Simulation results

We implemented the proposed model using the MATLAB tool with state-of-art techniques. We implemented all methods on the system equipped with Intel® Xeon® Gold 6130 processor, Central Processing Unit (CPU) of 2.10 GHz, RAM of 8 GB, and Graphical Processing Unit (GPU) of Titan RTX graphics card. We applied the proposed HDLA using different classifiers on both datasets and measure their performances in terms of accuracy, recall, precision, F1-score, and specificity parameters. These parameters are computed as:

$$accuracy = \frac{TP + TN}{TP + TN + FP + FN} \times 100 \quad (7)$$

$$precision = \frac{TP}{TP + FP} \times 100 \quad (8)$$

$$recall = \frac{TP}{TP + FN} \times 100 \quad (9)$$

$$specificity = \frac{TN}{TN + FP} \times 100 \quad (10)$$

$$f1score = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \times 100 \quad (11)$$

Table 5 C19RD dataset overall statistics (<https://www.kaggle.com/tawsifurrahman/covid19-radiography-database>)

“Number of chest X-ray samples	2905
Number of normal chest X-ray samples	1583
Number of Covid-19 chest X-ray samples	219
Number of viral pneumonia chest X-ray samples	1345
Total test samples (30%)	871
Total training samples (70%)	2034”

where, True Positive (TP) represents the correct inputs classified as correct, True Negative (TN) represents the incorrect inputs classified as incorrect, False Positive (FP) represents incorrect inputs classified as the correct, and False Negative (N) represents correct inputs classified as incorrect. We presented the simulation results in three sub-sections below.

4.1 CXIP dataset results

This section presents simulation results using the CXIP dataset consisting of three classes such as normal, bacterial pneumonia, and viral pneumonia. Figures 3, 4, 5, 6 and 7 show the outcome of detection accuracy, precision, recall, specificity, and F1-score metrics respectively. From these outcomes, it is observed that we have investigated the automatic CNN feature extractors and optimized CNN features using five different classifiers. The aim is to check the impact of applying the features scaling technique on automatically extracting CNN features on the performances of classifiers. We have conducted such investigations with the C19RD dataset also.

Figure 3 and its associated Table 6 show the detection accuracy performance of several classifiers. Figure 4 and Table 7 illustrate the performance of different classifiers in terms of accuracy rate. Figure 5 and Table 8 provide recall rate analysis using several classifiers. Figure 6 shows the specificity rate performance with its associated 9 for each classifier. Finally, Fig. 7 shows the F1-score rate with its associated 10 using different classifiers. We observed two things from these results: (1) the DNN classifier generated the highest lung disease classification performances among all classifiers, and (2) the optimized CNN features obtained performance gains over the raw CNN features.

The DNN classifier derived from the benefits of the deep learning algorithm with the inclusion of the fully connected layer, softmax layer, and classification layers with the total hidden layers set to 100 and the epoch rate set to 70. It has already been proven that deep

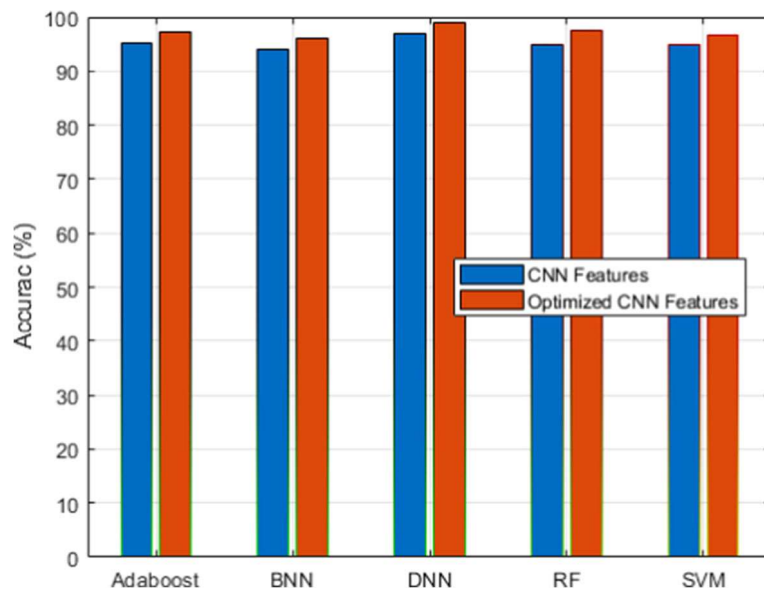


Fig. 3 Accuracy analysis for CXIP dataset

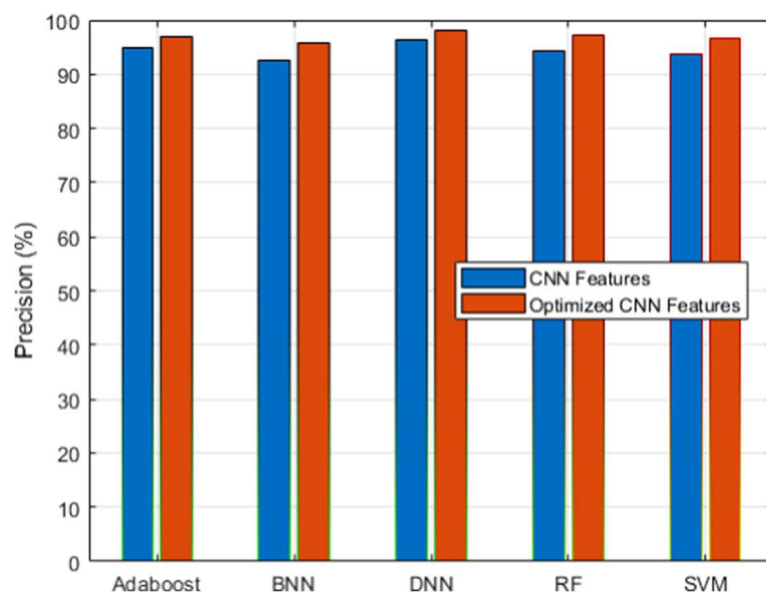


Fig. 4 Precision analysis for CXIP dataset

learning classifiers have the benefit of high recognition accuracy compared to conventional machine learning algorithms. Therefore, HDLA with DNN produced higher lung disease performances. Among other classifiers, ensemble classifiers such as Adaboost and RF delivered better performances than the SVM and BNN classifiers. On the other side, the optimized CNN features produced a higher performance than the original raw CNN features because of the scaling of CNN features. The scaling technique produced the features in a similar range by

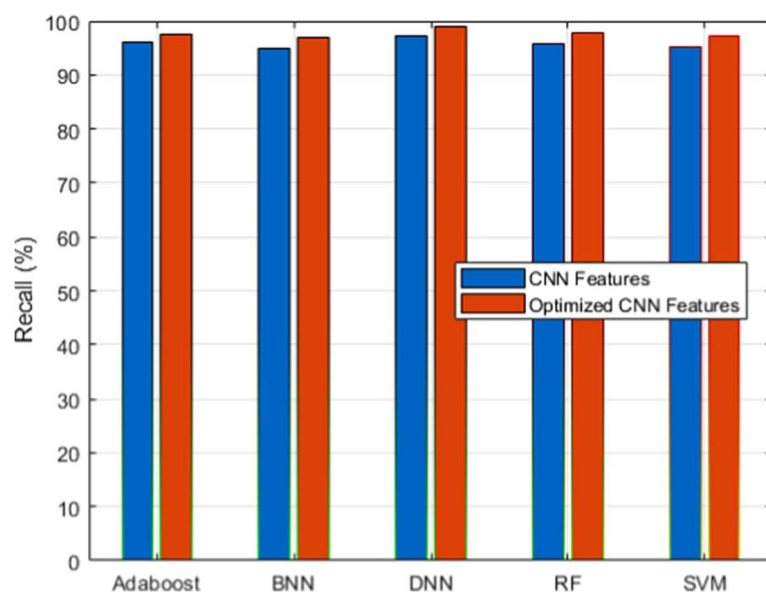


Fig. 5 Recall analysis for CXIP dataset

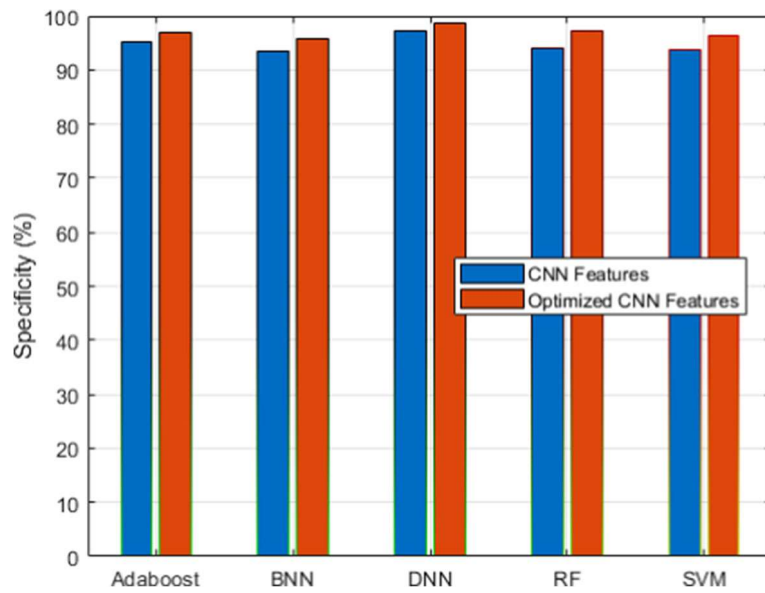


Fig. 6 Specificity analysis for CXIP dataset

discarding the variations that not only help to improve the performances of the classifiers but also reduce the overall training time. It is observed that feature scaling leads to performance improvement approximately by 2%.

For all the classifiers, the proposed HDLA model has produced improved classification performances consistently ranging from 95% to 99%. The key reasons for such improvements are the image quality enhancement without loss of data, a highly effective CNN features

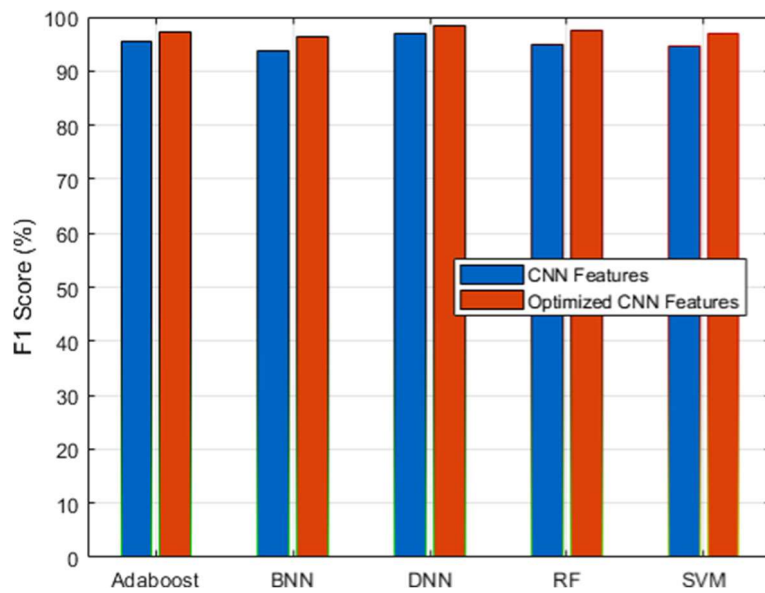


Fig. 7 F1-score analysis for CXIP dataset

Table 6 Performance of accuracy for CXIP dataset

Classifier	RF	SVM	BNN	Adaboost	DNN
CNN Features	94.99	94.89	93.88	95.13	96.95
Optimized CNN Features	97.61	96.78	96.11	97.29	98.99

Table 7 Performance of precision for CXIP dataset

Classifier	RF	SVM	BNN	Adaboost	DNN
CNN Features	94.18	93.78	92.56	94.88	96.34
Optimized CNN Features	97.23	96.57	95.78	96.91	98.12

extraction model using transfer learning and pre-trained ResNet50 model, and features scaling techniques. These performances improvement overcome the limitations of existing methods (Tables 9 and 10).

4.2 B. C19RD dataset results

We used a scalable C19RD dataset to test the scalability and reliability of the proposed approach. Figures 8, 9, 10, 11 and 12, together with their related Tables 11, 12, 13, 14 and 15, show the performance analysis of lung disease classification using the C19RD dataset in terms of accuracy, precision, recall, specificity, and F1-score parameters. This dataset includes roughly 2905 chest X-ray samples divided into three categories: normal, Covid-19 pneumonia, and viral pneumonia. With a few exceptions, all of the results indicate comparable performance patterns to those seen for the CXIP dataset. The total accuracy for the HDLA-DNN

Table 8 Performance of recall for CXIP dataset

Classifier	RF	SVM	BNN	Adaboost	DNN
CNN Features	95.88	95.31	94.98	96.11	97.34
Optimized CNN Features	97.87	97.12	96.98	97.41	98.95

Table 9 Performance of specificity for CXIP dataset

Classifier	RF	SVM	BNN	Adaboost	DNN
CNN Features	94.06	93.78	93.51	95.09	97.12
Optimized CNN Features	97.34	96.37	95.71	96.83	98.56

Table 10 Performance of F1-score for CXIP dataset

Classifier	RF	SVM	BNN	Adaboost	DNN
CNN Features	95.02	94.53	93.75	95.49	96.83
Optimized CNN Features	97.54	96.84	96.37	97.15	98.53

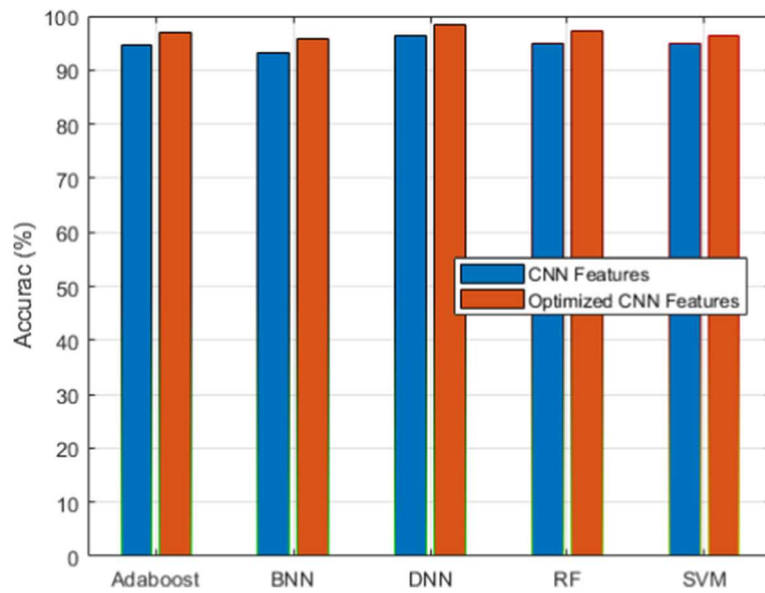


Fig. 8 Accuracy analysis for C19RD dataset

model is 98.99% when using CXIP and 98.35% when using C19RD. The arguments and findings surrounding the CXIP dataset's performance also apply to the C19RD dataset's performance.

Finally, we have presented the Receiver Operating Characteristic (ROC) curve in Fig. 13. From these results, it is indicated that the proposed using DNN provided a higher prediction accuracy compared to other classifiers.

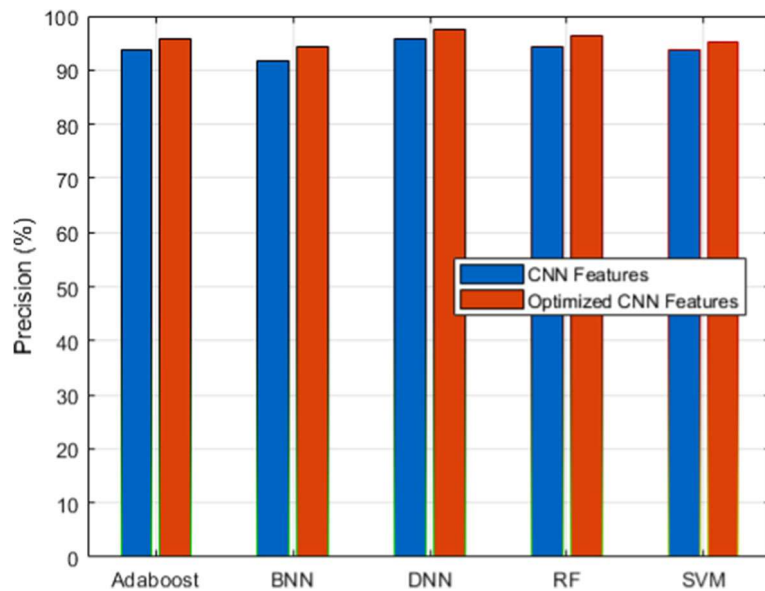


Fig. 9 Precision analysis for C19RD dataset

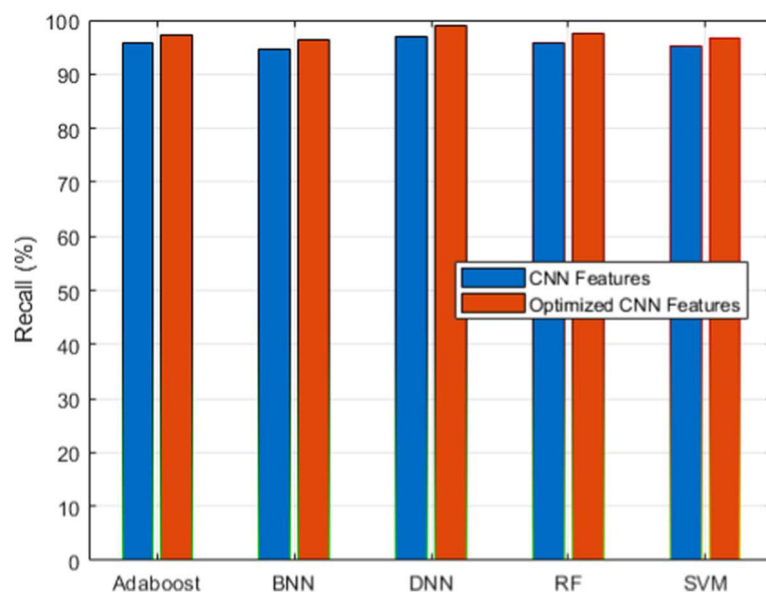


Fig. 10 Recall analysis for C19RD dataset

4.3 State-of-art analysis

Apart from the above results, this section presents a comparative study with similar methods those been recently proposed for lung disease classification using deep learning mechanisms. We have selected 6 such methods for the comparative study using both datasets in terms of overall accuracy, time complexity (training and testing average time), and space complexity.

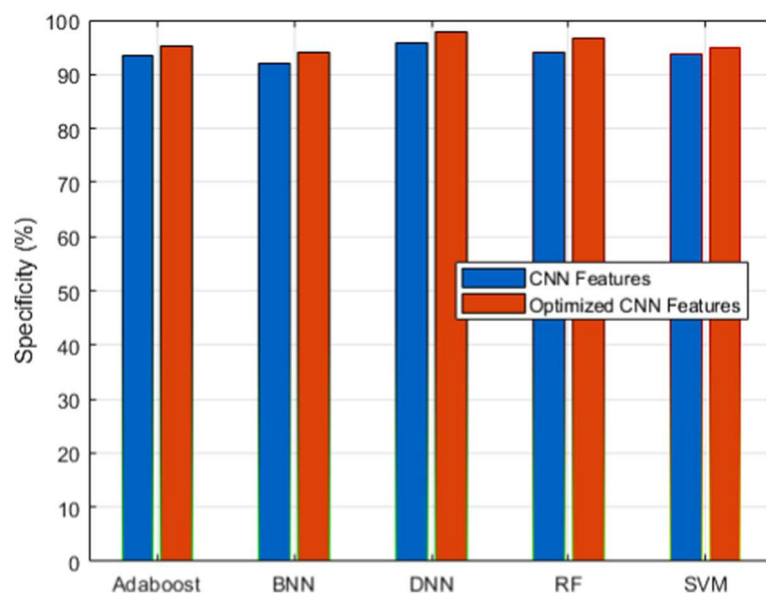


Fig. 11 Specificity analysis for C19RD dataset

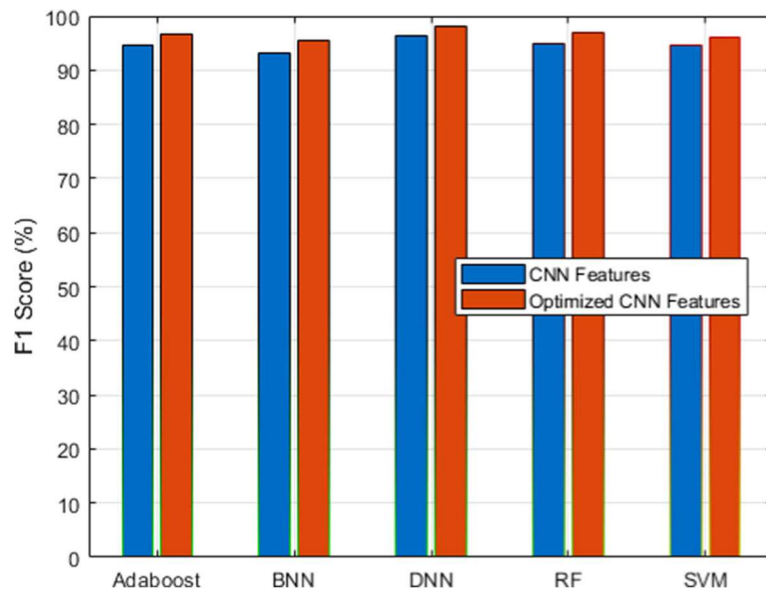


Fig. 12 F1-score analysis for C19RD dataset

The state-of-art methods such as CAAD [83], CheXGCN [20], CSEN [79], CNN-TL [58], VGG16-BM [57], and WE-CNNs [45] are implemented using CXIP and C19RD datasets under the similar experimental settings discussed above and 70:30% (training: testing) ratio. The functionality of these methods is closely related to the proposed model with the common goal of lung disease classification from chest X-ray images. We have already briefly reviewed these methods in section 2 of this paper. Tables 16 and 17 demonstrate the comparative analysis using CXIP and C19RD datasets respectively. From Tables 16 and 17 outcomes, the proposed model has outperformed all the recent state-of-art techniques in terms of efficiency and robustness metrics. The pre-processing of chest X-ray images, robust design of the CNN model, and feature scaling phases of the proposed HDLA model boost the classification performance with minimum computational efforts.

Table 11 Performance of accuracy for C19RD dataset

Classifier	RF	SVM	BNN	Adaboost	DNN
CNN Features	94.99	94.89	93.20	94.51	96.31
Normalized CNN Features	97.11	96.21	95.72	96.81	98.35

Table 12 Performance of precision for C19RD dataset

Classifier	RF	SVM	BNN	Adaboost	DNN
CNN Features	94.18	93.78	91.66	93.78	95.71
Normalized CNN Features	96.37	95.12	94.43	95.73	97.43

Table 13 Performance of recall for C19RD dataset

Classifier	RF	SVM	BNN	Adaboost	DNN
CNN Features	95.88	95.31	94.57	95.63	96.94
Normalized CNN Features	97.51	96.74	96.32	97.34	98.87

Table 14 Performance of specificity for C19RD dataset

Classifier	RF	SVM	BNN	Adaboost	DNN
CNN Features	94.06	93.78	91.91	93.44	95.78
Normalized CNN Features	96.59	94.89	93.89	95.31	97.67

Table 15 Performance of F1-score for C19RD dataset

Classifier	RF	SVM	BNN	Adaboost	DNN
CNN Features	95.02	94.53	93.10	94.69	96.32
Normalized CNN Features	96.93	95.92	95.36	96.52	98.14

The outcomes reveal that the proposed model will further be required to integrate with smart healthcare systems for remote health monitoring as mentioned in recently proposed studies [39, 40, 55, 63].

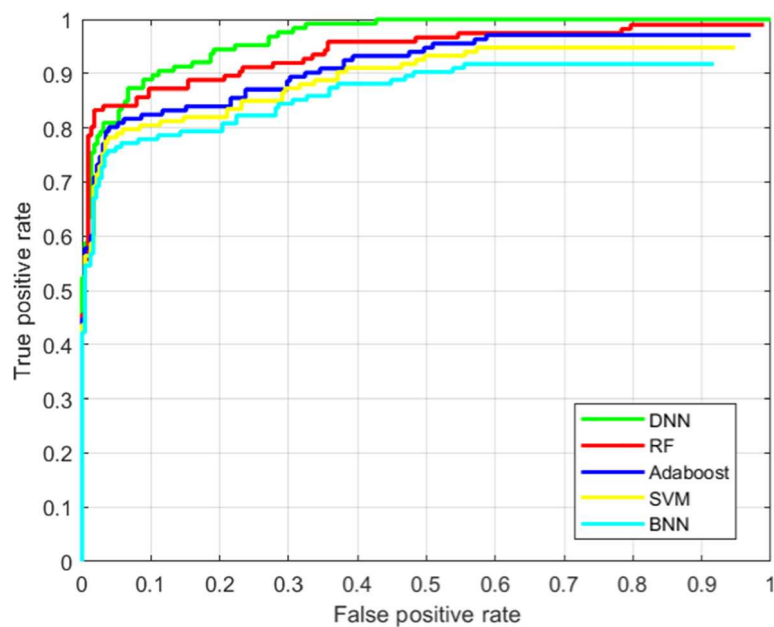
**Fig. 13** ROC performance analysis of the proposed model

Table 16 State-of-art methods comparative study using CXIP dataset

Methods	Year	Overall Accuracy (%)	Time Complexity (Seconds)	Space Complexity (Bytes)
CAAD [83]	2021	91.34	4788	859
CheXGCN [20]	2020	96.62	5671	1132
CSEN [79]	2021	97.12	5389	1029
CNN-TL [58]	2020	95.25	4656	805
VGG16-BM [57]	2020	93.19	5383	1011
WE-CNNs [45]	2021	96.87	6488	1378
HDLA-DNN	Proposed	98.99	4367	633

4.4 Limitations

Although the proposed model delivered higher accuracy in lower computational requirements compared to existing techniques but suffered from a few limitations including:

- The severity analysis of disease is yet another vital requirement for the appropriate medical treatment which is missing in the proposed model.
- The proposed model failed to estimate the Region of Interest (ROI) in the input chest X-ray images.
- Lack of real-time X-ray image analysis using the proposed model.

5 Conclusion and future work

We proposed the novel framework HDLA of lung disease detection using deep learning and machine learning techniques from the chest X-ray images. The framework was built via the steps such as pre-processing, robust features extraction and scaling, and classification. The pre-processing step produced the improved quality of chest X-ray images to boost the feature extraction and classification performances. The robust CNN model had designed using the pre-trained model for the 2D input image and produced the 1D feature vector with optimum kernel design. To reduce the processing speed and the memory requirements, we applied the features scaling mechanism integrated with the proposed 2D CNN model. To detect lung disease from the input chest X-ray images, we employed the different underlying classifiers. The hybrid approach of the deep learning model for automatic feature extraction and classification overcomes the research problems of the state-of-art methods. The experimental outcomes

Table 17 State-of-art methods comparative study using C19RD dataset

Methods	Year	Overall Accuracy (%)	Time Complexity (Seconds)	Space Complexity (Bytes)
CAAD [83]	2021	91.02	3918	751
CheXGCN [20]	2020	96.21	4962	899
CSEN [79]	2021	96.53	4493	802
CNN-TL [58]	2020	94.93	3845	722
VGG16-BM [57]	2020	92.88	4482	798
WE-CNNs [45]	2021	96.52	5659	1129
HDLA-DNN	Proposed	98.35	3356	489

using two chest X-ray datasets revealed the efficiency of the proposed model over the state-of-art similar methods in terms of overall accuracy, processing speed, and memory requirements. The lung disease classification accuracy of the proposed model has improved by 3.1% and reduced the processing speed by 16.91% and space requirements by 15.6% compared to existing methods. Several interesting research directions to extend the functionality of HDLA include (1) technique to perform the lung disease severity analysis, (2) improving the HDLA model by introducing the sequential deep learning classifiers, (3) creating real-time pneumonia and Covid-19 chest X-ray datasets, and (4) applying handcrafted features for the severity analysis.

Data availability Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

Declarations

Ethical approval This article does not contain any studies with human participants performed by any of the authors.

Conflict of interest All authors declares that they has no conflict of interest.

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